



Dynamic scheduling mechanism for intelligent workshop with deep reinforcement learning method based on multi-agent system architecture

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ABSTRACT

With the development and changes of industry and market demand, the personalized customization production mode with small batch and multiple batches has gradually become a new production mode. This makes production environment become more complex and dynamic. However, traditional production workshops cannot effectively adapt to this environment. Combining with new technologies, transforming traditional workshops into intelligent workshop to cope with new production mode become an urgent problem. Therefore, this paper proposes a multi-agent manufacturing system based on IoT for intelligent workshop. Meanwhile, this paper takes flexible job shop scheduling problem (FJSP) as a specific production scenario and establishes relevant mathematics model. To build the agent in intelligent workshop, this paper proposes a data-based with combination of virtual and physical agent (DB-VPA) which has information layer, software layer and physical layer. Then, based on the manufacturing system, this paper designs a dynamic scheduling mechanism with deep reinforcement learning (DRL) for intelligent workshop. This method contains three aspects: (1) Modeling production process based on Markov decision process (MDP). (2) Designing communication mechanism for DB-VPA. (3) Designing scheduling model combining with improved genetic programming and proximal policy optimization (IGP-PPO) which is a DRL method. Finally, relevant experiments are executed in a prototype experiment platform. The experiments indicate that the proposed method has superiority and generality in solving scheduling problem with dynamic events.

1. Introduction

The manufacturing industry is the material base and industrial main body of the national economy and is essential to create social wealth and achieve sustainable development. Affected by the global economic integration and market diversification, the demand of customers for products tends to be diversified, personalized and dynamic. Personalized customization (multiple varieties, medium and small batches) has gradually become the new production mode, and the production process has gradually emerged high concurrency, mixed lines and difficult to predict. This makes the production environment of the manufacturing system increasingly full of uncertainty and often accompanied by continuously changing and unpredictable tasks (changes in production tasks, emergency tasks, etc.) and events (equipment failures, resource changes, etc.). Therefore, the deal with dynamic events and the flexibility required to process complex products at the organization and scheduling level. Moreover, the scheduling method of manufacturing system can effectively solve dynamic events in time and meet the

requirements of multiple objectives (such as time, energy, cost, flexibility, etc.). Therefore, current manufacturing system should consider two key issues which include building a reasonable manufacturing system architecture and designing a real-time scheduling method to effectively respond to the dynamic manufacturing environment and quickly adapt to various changing needs.

To make the manufacturing system better adapt to the new production mode, many new manufacturing system control and organization modes have been proposed by scholars, such as multi-agent manufacturing system, holon manufacturing system, fractal manufacturing system (Nie, Tang, Zhu, & Sun, 2022; Ni and Yan, 2022; Peralta and Soltero, 2020). Among them, multi-agent manufacturing system is the most widely studied. The multi-agent manufacturing system regards the equipment or function module as an agent. The agents can independently deal with various events through information interaction, negotiation and calculation, and realize distributed self-organization production. This real-time scheduling mode can respond to various changes in the manufacturing system in a timely manner.

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With the development of “Industry 4.0”, the new generation of communication and sensor technologies are rapidly and highly integrated with the manufacturing industry. Especially the application of Internet of Things technology and manufacturing technology (IoTM) pushes the manufacturing system to the direction of digital, information and intelligent development (Sigov, Ratkin, Ivanov, & Xu, 2022). The realization of IoT manufacturing based on multi-agent manufacturing system has natural advantages in processing personalized orders, eliminating disturbance and workshop reconstruction. Therefore, multi-agent system is an ideal way to realize the IoT manufacturing. Combined with multi-agent system and relevant advanced technologies, designing a standard architecture of intelligent workshop to realize the transformation, and upgrading of the workshop is a research hotspot and difficulty point.

The manufacturing system architecture is just to improve the production performance of the workshop from physical level. However, there are many heterogeneous production resources and different kinds of products in workshop. The workshop production resource allocation is a key point. Resource allocation is actually a difficult and complex workshop scheduling problem. Therefore, excellent production scheduling can effectively complex scheduling problems with some relevant goals (energy saving, shortening delivery time, etc.) (Li, Huang, Wu, & Guo, 2020) and make manufacturing system to meet the requirement of new market and production mode from software level. The production scheduling has been widely studied recently by many scholars. Flexible job shop scheduling problem (FJSP) can be regarded as an important and common workshop scheduling problems which can be seen a classic Nondeterministic Polynomial hard problem (Brandimarte, 1993) in the manufacturing field. The previous research on FJSP did not consider the transportation of AGV and assume the transportation resources are sufficient. Obviously, this does not meet the actual production needs when the job number is large. Therefore, the workshop should consider not only the scheduling of machines but also the scheduling of AGV. Moreover, the production environment is not a static state but a dynamic state. The manufacturing system will be distributed by some uncertain events (e.g., machine failure, job insertion, delivery date change). This requires the manufacturing system to respond in real time and make adjustments in time. The scheduling method should consider both resource allocation and uncertain events to achieve real-time scheduling.

In the past few years, many scholars adopted heuristic algorithm (scheduling rules) and meta-heuristic algorithm (Bahamian, 2017; Shen, Zhao, Xia, & Du, 2017) to solve production scheduling problem. The heuristic algorithm can generate a scheduling rule to solve scheduling problems. But a single scheduling rule cannot solve the scheduling problem in all scenarios especially the dynamic events. The meta-heuristic algorithm can generate high-quality scheduling rules through evolution and other means. However, this method will spend much time to build a scheduling scheme under the complex environments, which cannot realize the real-time scheduling. The reinforcement learning (RL) provides another direction to solve this problem. The decision process often be considered to model as a Markov decision process (MDP) in RL (Liang et al., 2021). In the MDP, agent can choose action according to the current state which is observed by themselves. Then, environment will execute the action and feedback rewards to agent. Agent can adjust the policy to choose action according to the reward. Through this method, agent will learn excellent policy to choose action, so as to maximize the expected rewards. The production process can be modeled as MDP and intelligent can be seen as the agent. The agent can choose suitable scheduling scheme in each scheduling moment according to the actual environment, which can deal with dynamic interference events in real-time. However, traditional RL cannot deal with the data of enormous action space in intelligent workshop. In order to address this problem, some scholars combine neural network with RL, and then propose a method of DRL.

As a result, a multi-agent manufacturing system based on IoT is

designed firstly. The agents in the multi-agent system should have the following functions: physical device interconnection, information perception and communication, self-decision-making. From these functions, this paper develops a data-based with combination of virtual and reality agent (DB-VPA) to build the multi-agents in intelligent workshop. Then, this paper takes FJSP considering limited transportation resource (FJSP-LTR) as an actual scheduling problem of intelligent workshop. And a communication mechanism is designed to realize the communication of each agent. Then, to make each agent have scheduling ability, we propose a dynamic scheduling mechanism and IGP-PPO which is a DRL method to build and train scheduling model. In detail, DRL mainly divided into value-based methods and policy-based methods. The PPO (proximal policy optimization) algorithm is a strategy-based approach that utilizes the actor critic framework to train agents. The the quality of traditional action space in DRL for production scheduling is not clear. In order to solve this problem, the IGP (improved genetic programming) is be used to generate high quality action space for agent.

The main contributions in this paper have four points: (1) A multi-agent system based on IoT is proposed. Moreover, this paper establishes a multi-objective problem (minimize makespan and energy consumption) model for FJSP-LTR. (2) The DB-VPA is developed to realize the building of agents in the intelligent workshop which contain three parts (information layer, software service layer and physical layer). (3) In order to realize individual self-decision-making and group cooperation, a communication mechanism and a dynamic scheduling mechanism are designed for FJSP-LTR, and the IGP-PPO method is proposed to finish the training of scheduling model. (4) The experimental platform is built to verify the performance of proposed method.

Based on the above, the remain contents is described as follows. Section 2 introduces the related research situation and development trends. In section 3, this paper proposes a multi-agent system manufacturing system based on IoT and formulate multi-objective problem model for FJSP-LTR to achieve the goal of minimizing makespan and energy consumption. Section 4 describes the design of DB-VPA and elaborates the structure and relationship between each functional layer, including information layer, software layer and physical layer. Modeling the production process of intelligent workshop which consider the FJSP-LTR as MDP is introduced in Section 5. It also describes the multi-agent communication mechanism and provides the implementation details of dynamic scheduling model based on IGP-PPO. In Section 6, the experiment and relevant results are provided. Lastly, Section 7 concludes the whole paper and gives the direction of future study.

2. Literature review

The research in this paper mainly includes the construction of multi-agent system and using DRL method to solve FJSP-LTR in intelligent workshop. Based on this, the section 2 briefly describes the manufacturing system with new technologies, and the study of relevant method in the field of production scheduling.

2.1. Manufacturing system

The manufacturing system is the hardware foundation of intelligent workshop. Recently, with the rapid development of new technologies, there are new opportunities and challenges for intelligent workshops. However, the application of new technologies in intelligent workshops should consider the improvement and upgrading of traditional manufacturing systems of workshop.

In the past research, scholars presented several manufacturing systems to control the production progress. As a well-known reference architecture, PROSA has been used in holonic manufacturing systems which can translate computer operating systems into MES. Then, through improving and refining PROSA to build the ARTI (Activity-Resource-Type-Instance), communication between different production

resources can be improved (Valckenaers, 2020). In order to realize the optimality and reactivity of dynamic- hybrid control system, Jimenez et al. (Jimenez, Bekrar, ZambranoRey, Trumeaux, & Leitão, 2017) proposed Pollux as a D-HCA that contains a switching mechanism to steer the operating modes of a control system architecture and to ensure an adequate configuration according to real-time needs. Under the background of Industry 4.0, many scholars are studying how to combine new technologies and concepts with workshop to improve the production performance. Zhang et al. (Zhang, Tang, Zhu, Wang, & Nie, 2023) introduced a novel multi-agent architecture-based dynamic control system for processing workshops that utilizes the concept of “processing cells.” These cells are created based on the workshop’s function and are responsible for different aspects of the manufacturing process. Liu et al. (Liu, Zhang, Zhang, Tao, & Wang, 2019) presented a multi-agent system aiming to schedule aggregated resources efficiently. This system is composed of two parts: a platform-level scheduling MAS and an enterprise-level scheduling MAS. Zhang et al. (Zhang, Tang, Liu, Malekian, & Li, 2019) developed a collaborative virtual manufacturing environment (VME) using a multi-agent-based approach. The VME aims to reduce energy consumption and enhance efficiency in the manufacturing process while achieving a high level of autonomy for the manufacturing system. Ocker et al. (Ocker, Kovalenko, Barton, Tilbury, & Vogel-Heuser, 2019) developed a method to connect resource agents with product agents by initializing each agent with a knowledge base (KB) that utilizes semantic web technologies (SWT), which can facilitate the connection between the agents. To address scheduling problem under the highly dynamic environment, Zhang et al. (Zhang, Sun, Liu, Zhang, Zhao, & Shen, 2021) developed a multi-agent system structure that emphasizes quick response times and utilizes simple communication networks. Leng et al. (Leng et al., 2022) designed an autonomous process control framework based on cloud-edge orchestration, which can realize efficient manufacturing of mass individualization. Kovalenko et al. (Kovalenko, Balta, Tilbury, & Barton, 2023) proposed a cooperation framework based on discrete events for the product agent to improve its flexibility, adaptability, and control strategy. The smart manufacturing system (SMS) proposed by Kim et al. (Kim, Lee, Son, Bae, & Chung, 2020) employs a MAS combined with RL. The SMS consists of machines equipped with intelligent agents which can realize the decision-making, interaction, and the self-learning. Based on cloud manufacturing, Liu et al. (Liu, Wang, Wang, Wang, & Zhang, 2018) presented a MAS model, which incorporates many-to-many negotiations. Under the background of mass personalization, Qin and Lu (Qin & Lu, 2021) presented a new industry network named self-organizing manufacturing network to adapt to new production models. Kovalenko et al. (Kovalenko, Tilbury, & Barton, 2019) developed a product agent architecture with the ability to explore the local environment and plan and schedule tasks for the manufacturing system. ElMaraghy et al. (ElMaraghy and ElMaraghy, 2022) proposed a novel paradigm called Adaptive Cognitive Manufacturing Systems (ACMS), which is characterized by cognitive adaptation. They also introduced the drivers, enablers, and other key features of this new paradigm.

2.2. Classic production scheduling method

The occurrence of dynamic events makes the intelligent workshop need to obtain real-time scheduling. Recently, scholars have proposed a lot of methods to solve this scheduling problems. They mainly conclude *meta*-heuristic method, hyper-heuristic method, and other novel methods.

The meta heuristic method is mainly based on the natural-like method which combine random algorithm and local search algorithm. Regarding the distributed flow-shop scheduling problem with blocking stage, Zhao et al. (Zhao, Bao, Wang, Xu, & Zhu, 2022) introduced a discrete heuristic and *meta*-heuristic approach that utilizes problem-specific knowledge. This method is capable of effectively solving this particular type of scheduling problem. Zheng et al. (Zheng, Zhang, Tian,

& He, 2021) designed an efficient hybrid *meta*-heuristic algorithm which combines three evolutionary algorithms to solve a flexible flow shop scheduling problem with tow actual constraints. Pan et al. (Pan, Gao, Li, & Wu, 2023) used five *meta*-heuristics to addresses a same problem named distributed lot-streaming permutation flow shop scheduling problem. However, *meta*-heuristic-based algorithms need a large number of state space and has a high dependence on assumptions about the data, which may lead to algorithm failure if the data does not meet expectations. Therefore, *meta*-heuristic-based algorithms cannot effectively find an excellent solution when the production is complex. Based on this, hyper-heuristic algorithm can manage or manipulate a series of low-level heuristic algorithms (LLH) to generate new heuristic algorithms and new scheduling rules. Therefore, compared with *meta*-heuristic-based algorithms, this method can fully utilize the advantages of various *meta*-heuristic algorithms to improve overall performance. Moreover, it also has adaptability and can dynamically adjust their strategies to adapt to changes in the problem. Li et al. (Li et al., 2022) adopted hyper-heuristic related to genetic algorithms, which incorporates seven low-level heuristics, to address scheduling problems under the uncertain environments. Qiao et al. (Qiao, Wu, He, Li, & Chen, 2022) developed an adaptive genetic algorithm to tackle two-stage hybrid flow-shop scheduling problems. This algorithm can dynamically adjust the probabilities of crossover and mutation operations based on the population diversity. Bouazza et al. (Bouazza, Salles, & Trentesaux, 2021) implemented a hyper-heuristic (HH) based approach to enable efficient switching between different scheduling rules to deal with scheduling problem under the product-driven control systems. Zhou et al. (Zhou, Yang, & Zheng, 2019) proposed three types of hyper-heuristic methods for obtaining machine assignment rules and job sequencing rules to address multi-objective dynamic flexible job shop scheduling problems. Shao et al. (Shao, Shao, & Pi, 2023) presented a novel programming model to formulate hybrid blocking flow-shop scheduling problem. Based on this, they designed a learning-based selection hyper-heuristic framework for solving this problem. Whether it is metaheuristic algorithms or hyperheuristic algorithms, they only focus on a specific type of problem and do not have universality. And they just focus on the final solution of scheduling problems while neglecting every step of the production process which result in no real-time performance. With the development of new technologies, some novel methods (machine learning, neural network, etc.) have been studied to improve the real time, adaptability, and versatility of scheduling method. Zhang et al. (Zhang et al., 2022) presented a two-stage method based on a convolutional neural network to solve the FJSP with machine breakdown. Tian et al. (Tian and Zhang, 2021) proposed a deep learning-based scheduling model to realize dynamic scheduling. Zhou et al. (Zhou et al., 2022) adopted a smart scheduler to reasonably allocate job and can effectively address unexcepted disturbances in smart manufacturing factories. Su et al. (Su, Huang, & Li, 2022) proposed a self-organizing neural scheduler (SoNS) based on long short-term memory (LSTM) to improve special constraints in scheduling. Gu et al. (Gu, Li, Tang, Wang, & Yuan, 2022) used a self-organizing mapping neural network and improved Q-learning to realize real-time scheduling.

2.3. DRL method for production scheduling

Reinforcement Learning (RL) is a subfield of machine learning with the goal of enabling agents to learn how to make decisions by interacting with the environment to maximize a specified objective function. In RL, agent can learn adaptive strategies to achieve long-term goals by observing the state of the environment, taking actions, and adjusting their behavior based on feedback, specifically in the form of rewards from the environment. Deep Reinforcement Learning (DRL) integrates deep learning and reinforcement learning methods to tackle challenges in traditional reinforcement learning posed by high-dimensional state and action spaces. Neural networks are used to approximate value functions or policies, enabling learning in large-scale and complex

environments. Recently, DRL which is a real time, adaptability and versatility method has been used in many fields such as smart grids, path planning and so on. Therefore, some scholars have applied DRL in production scheduling to realize dynamic scheduling.

Due to the advantages of DRL, using RDL method to solve production scheduling is able to adapt to more complex production and address various dynamic events more effectively and environments. Lei et al. (Lei et al., 2022) presented a DRL-based framework and a graph neural network to automatically learn a policy for solving FJSP in real-time. Chang et al. (Chang, Yu, Hu, He, & Yu, 2022) proposed adopted a double deep Q-Network algorithm and designed relevant penalty strategy to solve the dynamic flexible job shop scheduling problem when the arriving time of job is random. Combined with cloud-edge collaboration, Zhang et al. (Zhang et al., 2022) used a distributed dueling deep Q network to optimize the weighted tardiness of jobs in real-time production scheduling, which can can better allocate processing resources for workpieces. To address dynamic flexible job shop scheduling problem with insufficient transportation resources, Li et al. (Li et al., 2022) developed a hybrid deep Q network and designed a dynamic scheduling strategy to deal with dynamic events. Chen et al. (Chen et al., 2021) presented a DRL-based dynamic scheduling algorithm to solve the formulated MDP problem which can achieve dynamic resource

management in workshop. To solve the multi- objective FJSP with fuzzy processing time problem, Li et al. (Li et al., 2022) proposed a MOEA/D based on reinforcement learning which contains three parts. Hu et al. (Hu, Jia, He, Fu, & Liu, 2020) developed an adaptive DRL real-time scheduling approach with mixed rule to solve FJSP and realize the minimizing the makespan and total delay. Wang et al. (Wang et al., 2023) designed d a novel DRL algorithm combined with long short memory neural networks and TD Errors to address the dynamic scheduling problem in non-permutation flow-shop. In summary, the scheduling framework based on DRL can effectively solve various types of workshop dynamic scheduling problems.

Based on the above literatures, we can find some meaningful points: (1) Few studies have systematically elucidated the specific methods for transforming and upgrading traditional manufacturing systems into intelligent manufacturing workshops, which including the rebuilding of architecture and the improving of control methods. (2) Mots studies neglect the real-time connection between agents and physical equipment, and the research of agent's self-decision combining artificial intelligence technology is lacking. (3) The dynamic scheduling is an important part in manufacturing, and a small number of researchers designed dynamic scheduling mechanism based on DRL and multi-agent. Moreover, the real-time production data and scheduling action

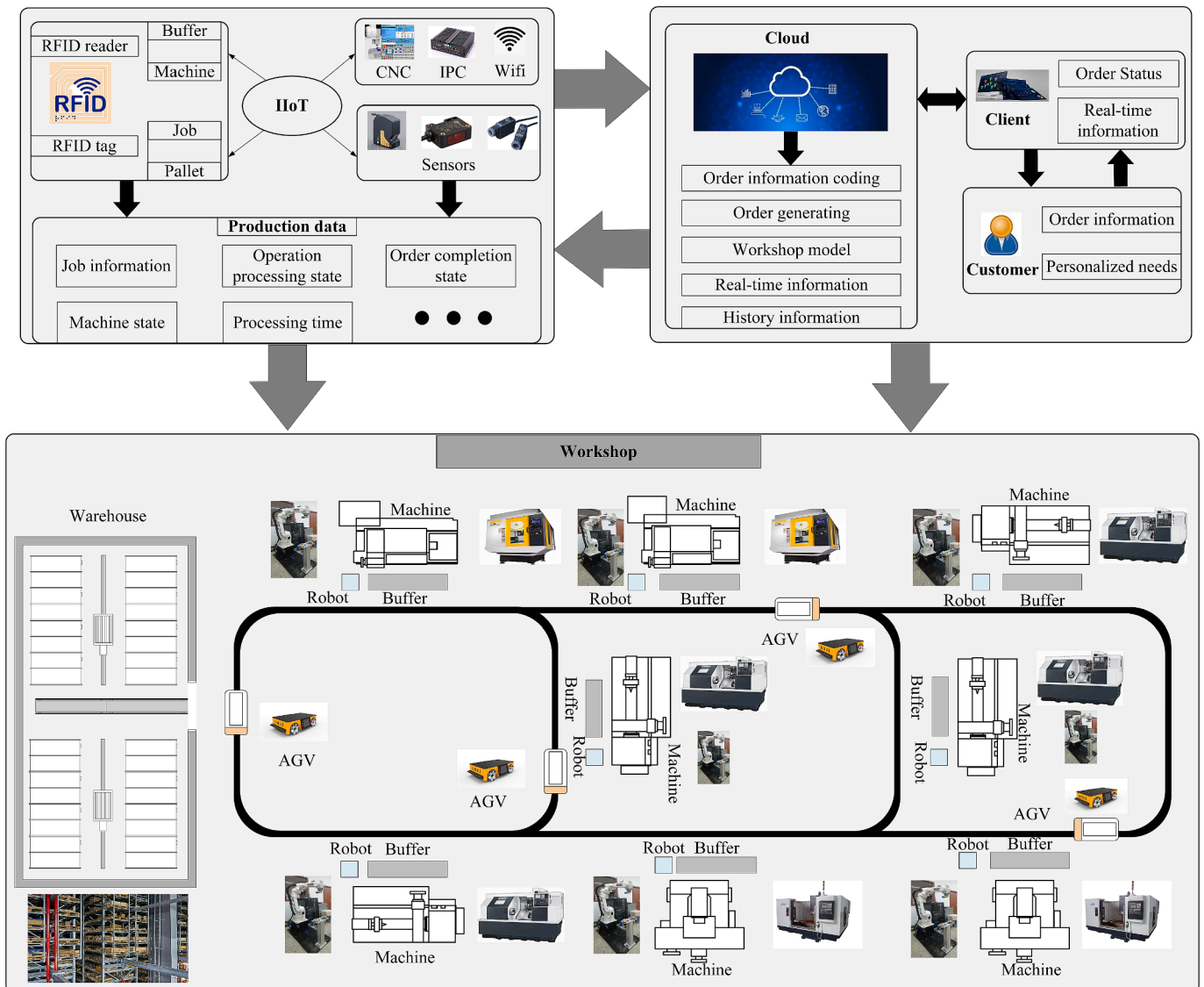


Fig. 1. Multi-agent manufacturing system based on IoT.

are not studied sufficiently. (4) In the above literatures, the scheduling problem for FJSP does not consider the situation of AGV transportation scheduling. Meanwhile, the scheduling goal of minimizing makespan and energy consumption is not studied.

3. Multi-agent manufacturing system based on IoT

3.1. The architecture of manufacturing system

The conventional architecture of manufacturing systems can no longer accommodate the demands of personalized production models. At the same time, the integration of advanced technologies has introduced new challenges and limitations to traditional workshops. To improve the manufacturing flexibility of intelligent workshop, this paper proposes a multi-agent manufacturing system based on IoT manufacturing which is shown in Fig. 1. The proposed architecture as shown in Fig. 1 has two characteristics: Firstly, cloud based and IoT technologies enable manufacturing systems to focus not only on the production process on the intelligent workshop, but also on the individual customization needs of customers. Secondly, this multi-agent manufacturing using edge computing can greatly improve the computational power and data processing of the system, and further improve the flexibility and reconfigurability of the intelligent workshop.

An order consists of multiple jobs with several operations. To complete an operation, a job needs to go through three stages: transportation, waiting, and processing. Based on this, this paper unitizes the production resources in the manufacturing system and constructs them into multiple groups of processing units. Among them, a group of processing units consists of the following parts: processing machines (CNC milling machines, machining centers), robot arm, buffers, industrial computers (IPC), sensors. Each group of processing units actually can be seen a process agent. And the processing unit also is a sensing node in the manufacturing system, an executor of processing production, and a decision maker of scheduling.

The adoption of advanced technology makes manufacturing system produce a large number of data which is high dimensional, redundant and dynamic. Meanwhile, the intelligent workshop has different kinds of heterogeneous equipment. Traditional agent-based manufacturing system cannot process complex data and realize the connection between heterogeneous equipment and IPC. Therefore, a novel design and implement of agent named data-based and combination of virtual and physical agent (DB-VPA) is designed to solve these problems. This proposed agent is an entity-oriented construction method to build different agents in manufacturing system.

3.2. Problem description

There are variable scheduling problems in industry. And the scheduling problem type of intelligent workshop studied in this paper is FJSP. To facilitate the study, this paper makes some assumptions about the processing process in the smart manufacturing shop: (1) each job must go through all processing operations; (2) each machine can only process one job at a given moment; (3) the loading and unloading time during processing are ignored, and each AGV has the same transport power; (4) an operation of a job can be processed on multiple machines, but the processing time is different; (5) the buffer between adjacent stages has a sufficiently large holding capacity. (6) All job does not need to be assembled, and the entire production process ends when all jobs are finished and returned to the warehouse.

The scheduling optimization objectives studied in this paper concludes minimizing makespan and energy consumption. The scheduling optimization objectives are formulated as follows:

$$T_m = \max_{1 \leq x \leq u} (FT_x) \quad (1)$$

$$NPT = \frac{T_m}{PT_u} \quad (2)$$

$$E_p = ME_p + TE_p \quad (3)$$

$$ME_p = \sum_{i=1}^m ((\sum_{x=1}^u \sum_{y=1}^p (BA_{iptxy} \cdot CP_{xyi}) + IP_{ipt} \cdot ICP_i) \cdot Time_p) \quad (4)$$

$$TE_p = \sum_{j=1}^n (TP_{jpt} \cdot TP \cdot Time_p) \quad (5)$$

$$\min G = \lambda_1 \cdot T_m + \lambda_2 \cdot E_p \quad (6)$$

$$FT_x \leq T_m, \quad x = 1, 2, \dots, u \quad (7)$$

$$Mi - \sum_{x=1}^u \sum_{y=1}^v A_{iptxy} \leq 0, \quad i = 1, 2, \dots, m; \quad pt = 1, 2, \dots, NPT \quad (8)$$

$$Mi - \sum_{i=1}^m BO_{xyi} \leq 0 \quad x = 1, 2, \dots, u; \quad y = 1, 2, \dots, v \quad (9)$$

$$\sum_{w=1}^m (BO_{x(y-1)w} \cdot PAT_{xyw}) \leq \sum_{j=1}^n \sum_{w=1}^m (BT_{jxy} \cdot BO_{xyz} \cdot LT_{jw}), \quad x = 1, 2, \dots, u; \quad y = 2, 3, \dots, v \quad (10)$$

$$\sum_{i=1}^m \left(A_{x(y-1)i} \cdot (ST_{x(y-1)i} + PT_{x(y-1)i}) + \sum_{w=1}^m BO_{xyw} D_i(PML_i, PML_w) \right) \leq \sum_{w=1}^m (BO_{xyw} \cdot PAT_{xyw}), \quad x = 1, 2, \dots, u; \quad y = 2, 3, \dots, v \quad (11)$$

$$\begin{aligned} & \sum_{y=1}^v \sum_{i=1}^m (BO_{xyi} \cdot PMT_{xyi}) + \sum_{i=1}^m (BO_{x1i} \cdot D_i(LU, PML_i)) \\ & + \sum_{y=2}^v \sum_{i=1}^m \sum_{z=1}^m (BO_{x(y-1)i} \cdot BO_{xyw} \cdot D_i(PML_i, PML_w)) + \sum_{i=1}^m (BO_{xyi} \cdot D_i(PML_i, LU)) \leq FT_x, \\ & x = 1, 2, \dots, u \end{aligned} \quad (12)$$

Where, Eq. (1) to Eq. (6) represents the scheduling optimization objectives. Eq. (1) represents the calculation of minimizing completion time and Eq. (2) to Eq. (5) represent the calculation of minimizing energy consumption. Eq. (6) represents the size constraint between two goals. Eq. (7) considers the relationship between the completion time and the completion time T_p of each job. Eq. (8) means that a machine can only process one operation at a time. Eq. (9) ensures that one operation can select only one machine for processing. Eq. (10) allows the AGV to remain in the machine position for a period of time until it receives a new transport task. Eq. (11) considers that there is a priority constraint between different operations of the same job. Eq. (12) represents that the workpiece JB_x will not be completed until all operations are completed and the corresponding entity returns to ASRS.

4. The design and implementation of DB-VPA

The previous research on agents focuses on the definition of internal modules and functional implementation of agents, and the realization of the corresponding functions through software simulation. Therefore, most of these studies ignore the physical devices that correspond to the agent. On the other hand, the intelligent manufacturing workshop will generate a large amount of data in the actual production process, and it will put a huge pressure on the computational power of the whole workshop if these data are processed centrally. This paper takes real-time information data into account in building agents. By this way, the computational pressure of the whole manufacturing system can be distributed to each agent which can greatly improve the computational power and data processing capability of the manufacturing system.

This paper proposes a data-based and combination of virtual and physical agent (DB-VPA). The overall structure of DB-VPA is shown in Fig. 2. DB-VPA can obtain real-time information about the current production state and monitor the real-time information through the software layer. According to this information, reasonable scheduling plan will be given to the physical layer, while the physical layer will feed the latest states to the software layer. This section describes the design and implementation of each functional layer of the proposed DB-VPA.

4.1. The function layer of DB-VPA

The corresponding functions of the DB-VPA include: (1) Information sensing and processing to generate high quality production data; (2) Interconnection with physical equipment to give control instructions and obtain equipment information; (3) Real-time coordination and rational allocation of production resources in the workshop based on local and global information of the agent, thus achieving scheduling goals (minimizing makespan and energy consumption). Based on the above contents, the DB-VPA presented in this paper is divided into three

major layers, including: information layer, software layer, and physical layer. Among them, the software service layer includes AI layer, communication layer, and conversion layer. The detail of each function layer of DB-VPA is shown in Fig. 3.

4.2. Information layer

During the manufacturing production, the intelligent workshop accumulates a large amount of real-time information, which is usually high-dimensional and redundant. In this paper, the real-time information data of the workshop from the actual production is divided into job information, machine information and AGV information. For job information, there are three main components: process information, operation information, and target information. The process information includes: the type of job, the number of jobs, and the process route; the operation information includes: the total number of operations, the available machines for the operation, and the completed operation; and the target information includes: the time consumption for each process, and the energy consumption for each process. Process information and operation information can be downloaded directly from the cloud and shared with the job layer. Machine information includes status information, production information, and target information. Among them, state information refers to the processing status of the machine (processing, waiting, fault); processing information refers to the process that the machine is currently processing and has processed; the target information here is the same as the target information in the job information above, which records the processing time and energy consumption of the manufacturing system in real time. And the information of AGV include number of AGVs, the current location, the transportation power and time. Based on this, this paper designs 26 production data which can be seen in Table 2.

The information layer mainly classifies and processes the acquired information to finally obtain useful production data. The mathematical models corresponding to these objectives are compiled into modules and stored in the information layer. Therefore, the information data layer can be well integrated with the information obtained from intelligent workshop in real time, which will be used to build and update the production environment. It can also be used in the AI layer for the rational allocation of resources in the smart shop floor through the coordination mechanism of multi agents.

4.3. Software layer

The software layer includes AI layer, communication layer, and conversion layer. It enables communication interaction and cooperation between data-based and combination of virtual and physical agents (DB-VPA) as well as connection to the physical and information layer. The software service layer is the most central functional layer of the DB-VPA.

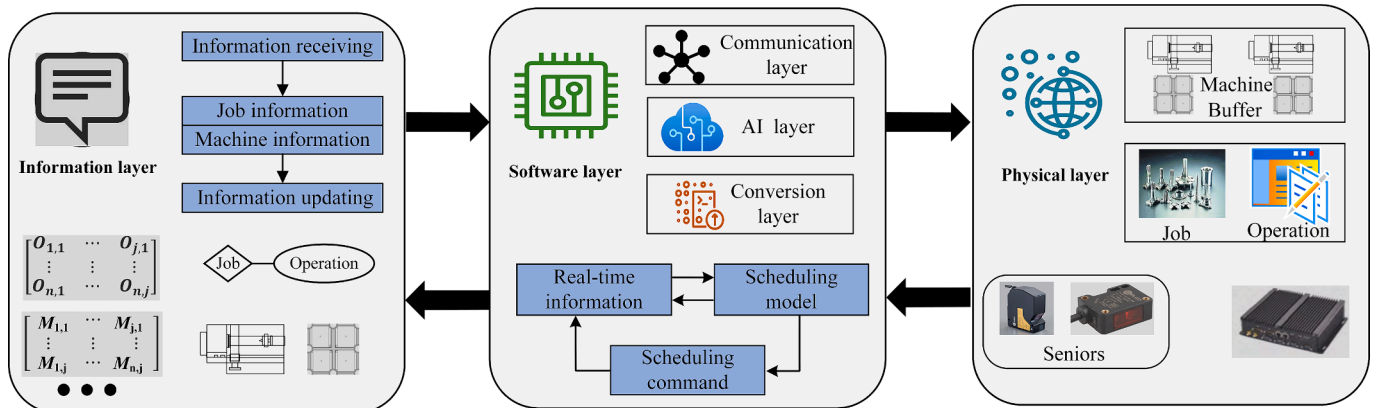


Fig. 2. The overall structure of DB-VPA.

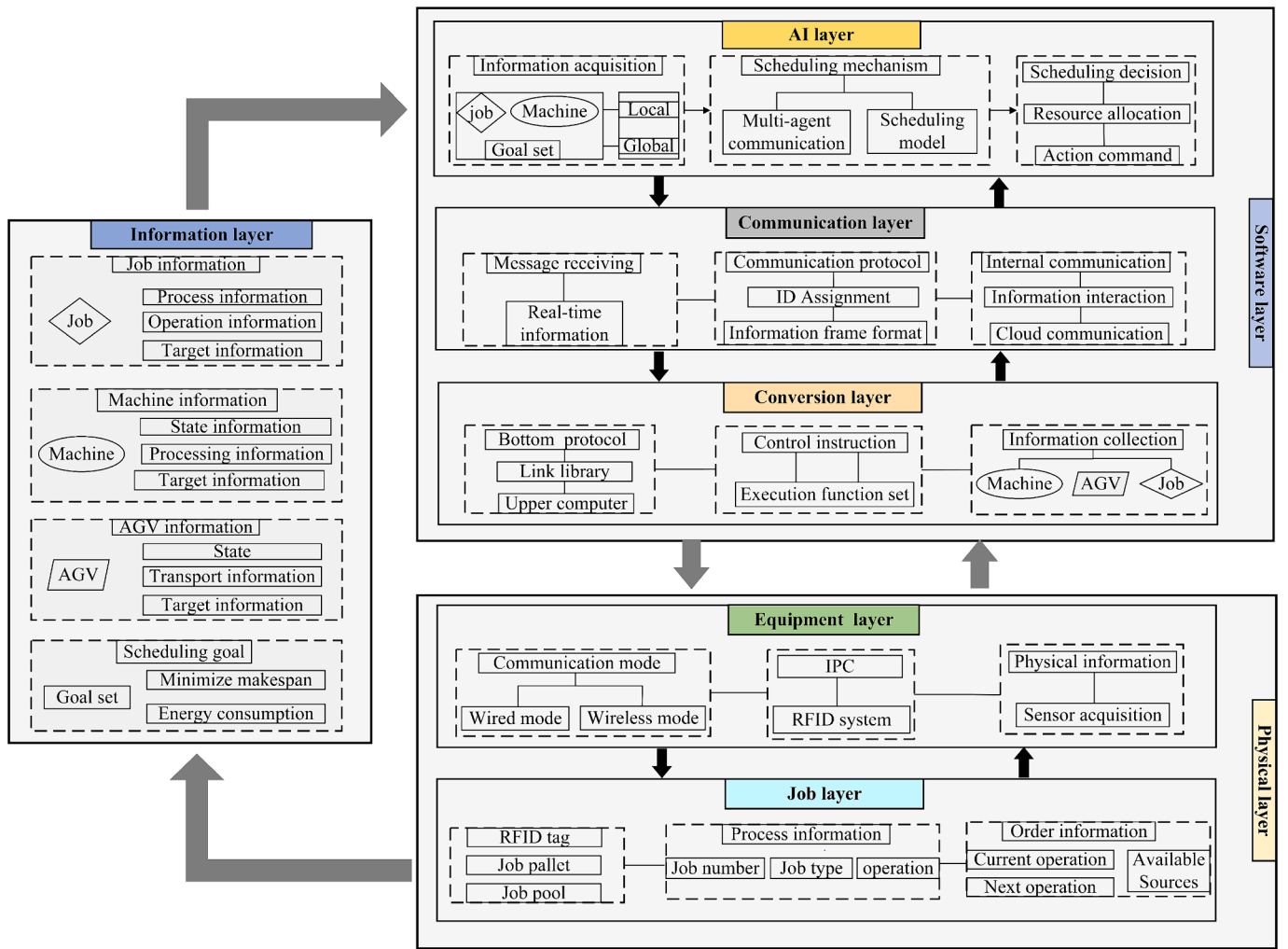


Fig. 3. The function layer of DB-VPA.

The software functional layer exists in the form of software programming in the industrial computer of the physical device.

4.3.1. AI layer

The AI layer is a real-time monitoring of the entire production process of the intelligent workshop. The information acquisition of AI layer has two parts. The first is the local information observed by themselves, and the second is the global information obtained from the information layer. Meanwhile, AI layer is also a decision maker which can be seen the brain of whole agent. Once there is a moment or disturbance event that needs to be scheduled, the AI layer can reasonably allocate jobs, processing machines and AGVs through the the scheduling model to achieve the goal of scheduling or effectively solve the disturbance event.

The scheduling processing in AI layer can be described as follow. First, the production state in workshop has been acquired and stored in information layer. This information will be converted into some matrix. AI layer can read this information matrix and input them into network model. Then, the network model can provide the most appropriate scheduling scheme to allocate processing job, machines and AGVs. Lastly, intelligent workshop will execute the scheduling command. It is noticed that the network model cannot used directly without training. This paper adopted the PPO algorithm to train the scheduling model which can be seen in Fig. 4. There are two neural networks in network model. The first network which is called actor network (AN) is used to choose action (scheduling scheme). The neural network 2 called critic network (CN) will evaluate the value of action choosing by AN and

calculate the advantage function. The AN will update the parameters according to loss function. When the training is finished, the AI layer will can be used to solve the scheduling problem.

4.3.2. Communication layer

The communication layer will consider three main aspects of communication of DB-VPA: the internal communication in the DB-VPA, information interaction between the DB-VPAs, and communication between DB-VPAs and cloud. The internal communication in DB-VPA include the input and output information matrix and format of each functional layer. The information interaction between the DB-VPA has three parts which include local information forming global information, communication protocols. Communication between the agents and the cloud should realize the function of obtaining order information and real-time customer requirements.

In order to facilitate communication and interaction between different DB-VPAs, this paper designs a scalable communication protocol, which can realize the real-time communication and information exchange between agents. Specifically, all equipment in the workshop has network communication function by embedding IPC, and the information frame format of communication between agents must meet the requirements of communication layer protocol. The proposed protocol utilizes object-oriented principles to obtain the features of physical equipment in the intelligent workshop. The physical equipment (each machine, each AGV and ASRS) can be seen an object in the workshop. And the object has three parts which contain class, state, and behavior.

Table 1

Notions used in intelligent workshop.

Notions	Description
x, xx	Index of jobs
y	Index of operations for a job
i, w	Index of processing machines
j	Index of AGV for transportation
pt	Index of unit time
PM	Collection of all machines, $PM = \{PM_1, PM_2, \dots, PM_m\}$
AM	Collection of all AGVs, $AM = \{AM_1, AM_2, \dots, AM_n\}$
JB	Collection of all jobs, $JB = \{J_1, J_2, \dots, J_u\}$
JO_{xy}	Collection of all operations of JB_x , $JO_x = \{JO_{x1}, JO_{x2}, \dots, JO_{xy}\}$
FT_x	The finishing time of JB_x
BA_{ipbcy}	0–1 variable that is 1 if JO_{xy} is processed on PM_i in pt , and 0 otherwise
ID_{ipt}	0–1 variable that is 1 if PM_i is idle in pt , and 0 otherwise
TP_{jpt}	0–1 variable that is 1 if AM_j is working in pt , and 0 otherwise
BO_{xyi}	0–1 variable that is set to 1 if operation JO_{xy} is to be processed on machine PM_i , and 0 otherwise
BT_{jxy}	0–1 variable that is 1 if JO_{xy} is to be transported by AM_j , and 0 otherwise
LT_{ji}	Time point when AGV AM_j leaves the machine PM_i
$Dt(a, b)$	Time required for AGV AM_j to move from position a to b
PML_i	The current location of PM_i in intelligent workshop
PAT_{xyi}	Time of JO_{xy} arrives PM_i
PST_{xyi}	Time of JO_{xy} starts processing in PM_i
PMT_{xyi}	Processing time of JO_{xy} on PM_i
LS	The location of warehouse
E_p	Total energy consumption of the workshop per unit time pt
ME_p	Total energy consumption of all machines per unit time pt
TE_p	Total energy consumption of all AGVs per unit time pt
CP_{xyi}	Cutting power of operation JO_{xy} on machine PM_i
ICP_i	No-load power of machine PM_i
TP	Transporting power of AGV
PT_u	Length of each unit of time
NPT	The number of units of time in the completion time
T_m	Makespan
CE_{total}	Total energy consumption
Mi	A very small constant less than 1

Table 2

The production data of intelligent workshop.

Production type	Data type					
	Static data	Dynamic data				
Job	Job number	1	Finished job number	5	Finished operation	9
	Job type	2	Processing job number	6	Processing operation	10
	Operation number	3	Location of job	7	Processing time	11
	Operation type	4	Processing power	8		
Machine	Machine number	12	State 1 processing	15	Available machine number	18
	Machine type	13	State 2 waiting	16	Processing time	19
	Location	14	State 3 fault	17	Energy consumption	20
AGV	AGV number	21	Location	23	Transporting time	25
	Transport power	22	Transporting job	24	Energy consumption	26

The same type objects can be seen as a class, so as to realize information transmission. The specific ID allocation of object and class can be seen in Table 3. The state obtains the basic production information of an object. The behavior stores the changing command of the object. Based on the above description, the communication protocol adopts a unified information frame format which includes frame header, target equipment ID, source equipment ID, length, command, and check bit. And the information frame format can be seen in Fig. 5.

4.3.3. Conversion layer

The conversion layer enables the interconnection between the host computer and the devices. There are different types and brands of heterogeneous devices in the manufacturing system. Most of them have different underlying protocols and require different communication protocols. Therefore, it is very difficult to obtain information such as real-time processing status of the entities devices and to realize the control of the processing devices. Therefore, we build a conversion layer which develops a link library and a web interface for real-time access. The conversion layer can realize real-time interconnection between physical layer and software layer to satisfy the functions of action control and real-time information acquisition.

4.4. Physical layer

The above-mentioned content builds the DB-VPAs (software layer as well as information layer) from a virtual perspective. Considering only the virtual layer would lead to a weaker association between the agent and the physical device, which would not enable real-time control and action state acquisition. Therefore, this chapter presents a physical layer for the DB-VPA which includes equipment layer and job layer.

4.4.1. Equipment layer

The object of the equipment orientation is the heterogeneous equipment in intelligent workshop. The communication mode of the physical equipment should be determined firstly. This paper adopts two communication modes which are wired mode and wireless mode. The wired mode is used to realize the connection between the equipment and the sensor, which can obtain relevant production data through the sensor. Wireless mode can realize the communication connection with the industrial computer. Then, in order to monitor and control the process of the machine in real time, this paper defined a processing unit which obtain five parts. The five parts include: robot arm, processing machine, buffer, sensor, IPC. The robot arm is used to transport job from AGV to buffer and buffer to machine. The processing machine can process the operation of job. The buffer can store job temporarily if machine is working or the job is waiting for transporting to another processing unit by AGV. The sensor can collect the data of equipment and transmit this information. IPC controls the whole processing unit as the upper computer.

4.4.2. Job layer

The job flows in real time during process of production, so the state of the order is changed in real time. The job layer uses RFID system to track the processing state and changes in processing attributes of the job in real time. The RFID system adopts RFID tag to record the current processing operation, the next operation, processing machine and other information. The new state of job will be written into RFID tag after finishing an operation. The information of job state will be transmitted to the information layer for certain data processing. Then the AI layer will give a scheduling scheme according to the information.

The process of recoding and updating job information can be described as follows. When a new order arrive workshop, the storage will pick up the information from cloud and send raw material of jobs to process. Each job will be deposit on a pallet which attaches a RFID tag. The initial information will be written into RFID tag by the RFID reader of storage. Then, job will be sent to processing unit for processing by AGV. After job arrives the processing unit, the robot arm will place the job on the buffer. Then, the RFID reader reads the RFID tag of this job and send the information to processing machine and IPC. After finishing the current operation, the RFID reader will update the information in the RFID tag and the AGV will transport job to another processing unit to process next operation.

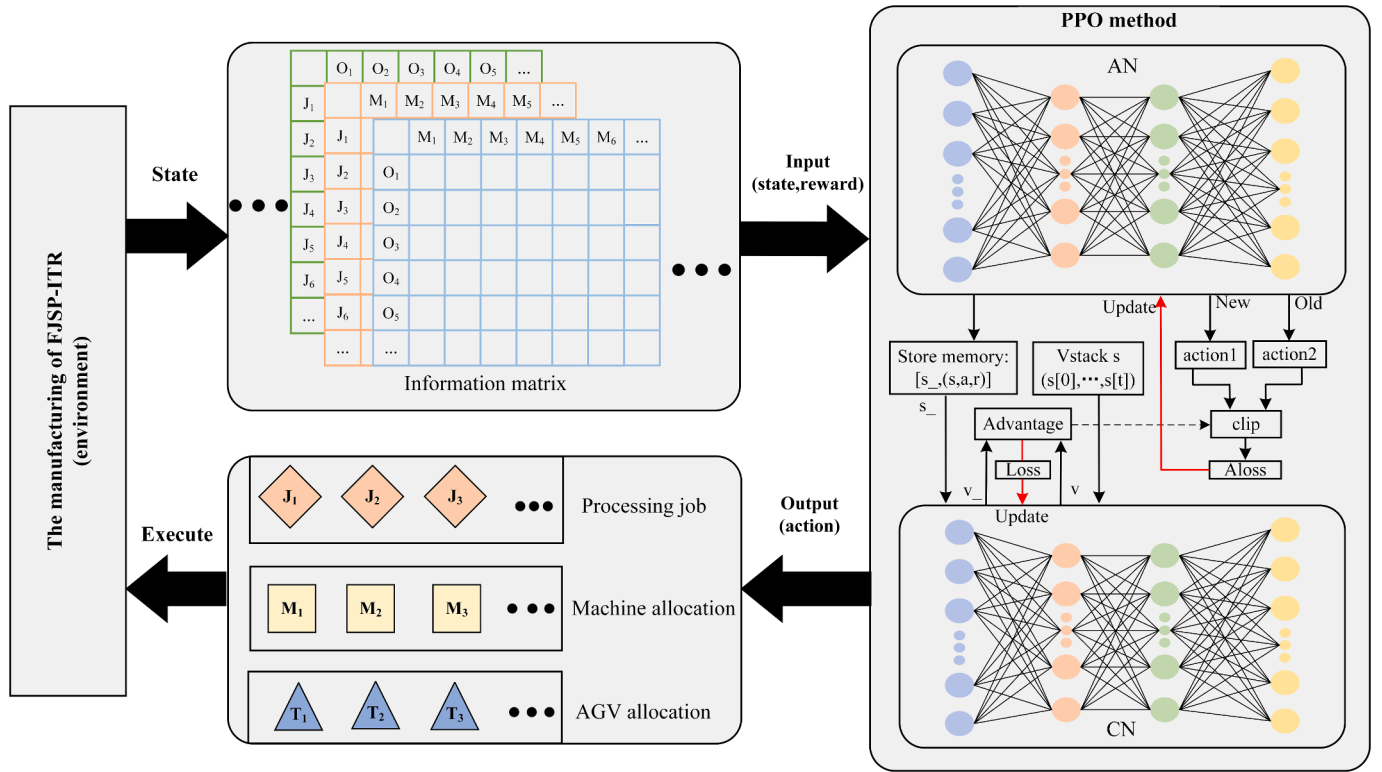


Fig. 4. Scheduling model in AI layer.

Table 3
The object and its ID in intelligent workshop.

(a)						
Object name	Storage	Machine	Robot arm	AGV	Sensor	Extending
Object ID	0x01-0xFF	0x01-0xFF	0x01-0xFF	0x01-0xFF	0x01-0xFF	...
(b). The class and its ID in intelligent workshop						
Class name	Storage	Machine	Robot arm	AGV	Sensor	Extending
Class ID	0x01	0x02	0x04	0x06	0x08	...

5. Dynamic scheduling mechanism

There are some agents in the intelligent workshop. These agents can be divided into storage agents, transport agents, processing agents, cloud agents. This paper proposed a concept of DB-VPA which will be used to realize the building of these agents. The DB-VPA can realize the functions of individual information perception, communication interaction and independent decision-making. However, the environment and actual production of intelligent is complex and dynamic which will bring some scheduling problem. Therefore, this section will introduce a dynamic scheduling mechanism based on DRL to solve complex and dynamic scheduling problems.

5.1. Environment building

5.1.1. The description of production environment

In order to get closer to the actual production process, this paper selects the FJSP which is common on the industry as the research object. Meanwhile, this paper also considers an actual case in the production of FJSP-LTR. The LTR means the number of AGV in workshop is not enough. After the job is processed, there is not enough AGV to transport it. The production process of FJSP is shown in Fig. 6.

The production of FJSP can be described as follows. At the beginning of the production process, raw material of all jobs is stored in the storage for waiting processing. In the scheduling process, the state of job is continuously changing which includes processing job, scheduled job, finishing job. Specifically, when the operation of the job needs to allocate a machine to process, or one of operation of the job has just finished processing and its next operation needs to allocate the machine. If the job meets the two situations, this job can be called processing job. When an operation of the job has been allocated to the machine but has not completed the processing, the job in this state can be called scheduled job. If a job has finished its all-own operation, the job in this state can be called finishing job. The finishing job will be stored in the storage lastly.

5.1.2. The building of environmental model

Based on the above description about production process, the intelligent workshop will generate processing job continuously after the

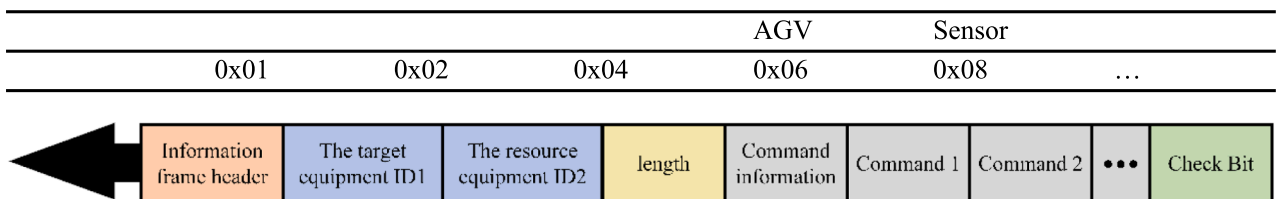


Fig. 5. Information frame format.

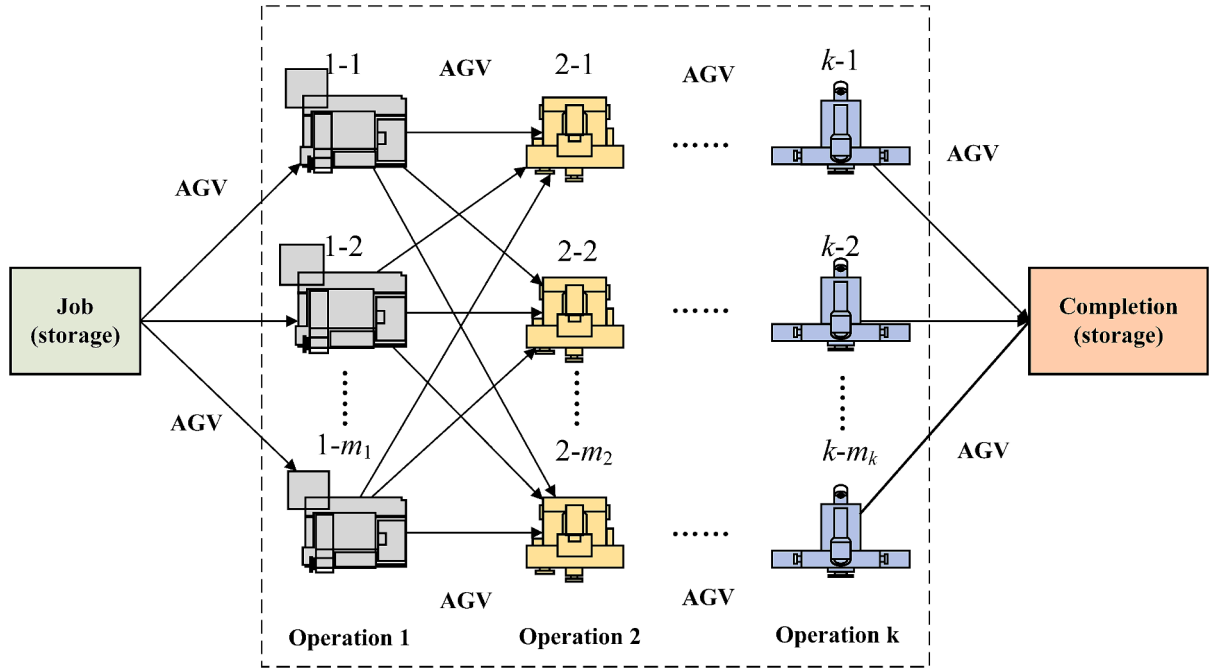


Fig. 6. The whole production process of FJSP-LTR.

order is released to the workshop. The DB-VPAs existed in the intelligent workshop should give an excellent scheduling scheme for processing job by scheduling method. The scheduling for intelligent workshop presents a time-progressive state. Specifically, whenever there is one or more jobs to be processed in the workshop, the DB-VPAs distributed in the workshop can obtain information of the current intelligent workshop and make a reasonable scheduling decision. Therefore, this process can be abstracted as a Markov decision process (MDP). The MDP usually consists of a quadruple which can be described as $E = \{S, A, P, R\}$. In the MDP, the machine that can learn and is responsible for making decisions is called the agent. Things that interact with agents are called environment E . Each agent can collect information which is used to learn from E and give command to change the state of E . The agent will observe environment state S and according to the strategy π to choose the action A . The action A will be executed by the environment. The state transition function P represents the possibility state changing of environment. When an action is executed, the agent will receive a reward according to the reward function R .

Based on this, the process production in the intelligent workshop can be molded an MDP. The model can be described as follows. The expression of environment model is M_e where consists of six elements, $M_e = \{N, S, A, T, R, \xi\}$. The element N represents the number of DP-VAs in the intelligent workshop. The station S is the production data of workshop which has been introduced in the above content. The action A represents the scheduling scheme which stored in scheduling library. The state transition function T represents the probability of environmental transfer. The reward function R represents the performance of scheduling goal and make agent to learn in the right direction. The ξ is the event collection which contain two cases. The first case is the event when the workshop exits job to be scheduled. The second case is the event when the production occurs dynamic disturbance events, including order inserting and machine failure.

The production in the intelligent workshop is carried out according to the time step if the process of production is normal. When the case (more than one) in event ξ is occurred, the whole workshop will schedule according to state S . The state S contain job information and equipment information of each DB-VPA. After scheduling, current jobs to be processed will be given processing priority and assigned to the appropriate machine for processing.

5.2. Communication mechanism

The communication between agents is very important during the production. And communication mechanism between agents is the basis of manufacturing data transmission. To realize the communication between agents, we designed a novel dynamic communication mechanism which contains six steps. And the process of communication mechanism can be seen in Fig. 7.

(1) The first step is information initialization stage. After the order arrive workshop from cloud, the agents in the workshop will be registered in the cloud (transmit the information) and finish the operation of initialization. The the target mathematical model of scheduling and environment model will be built and generated in the AI layer of each agent (DP-VA). Then, the workshop has ready to begin process.

(2) The second step is requesting scheduling stage. When an event at some time point belong to the event collection, the scheduling will be requested. The intelligent workshop will request scheduling and inform the relevant agents. If there are no scheduling requests, the whole production process will be normal.

(3) The third step is scheduling information transmitting stage. After receiving the scheduling request, the agents in workshop and job will transmit the information for scheduling. After receiving information, the AI layer will deal this information for scheduling.

(4) The fourth step is the stage of scheduling command execution. Job will be assigned to the machine and AGV for processing and transporting according to job processing priority. The production information of agents will be continuously updated during this stage. This information will be uploaded to the cloud for supervising.

(5) Repeat execution stage is the fifth step. After finishing scheduling command, whole production will process normal. If new scheduling event occurs, the step (2), (3), (4) will be repeated.

(6) The six step is finishing stage. When the production is over, all jobs have been processed and transported to the warehouse. The scheduling model in AI layer will analyze and summarize the scheduling performance and adjust its own parameters.

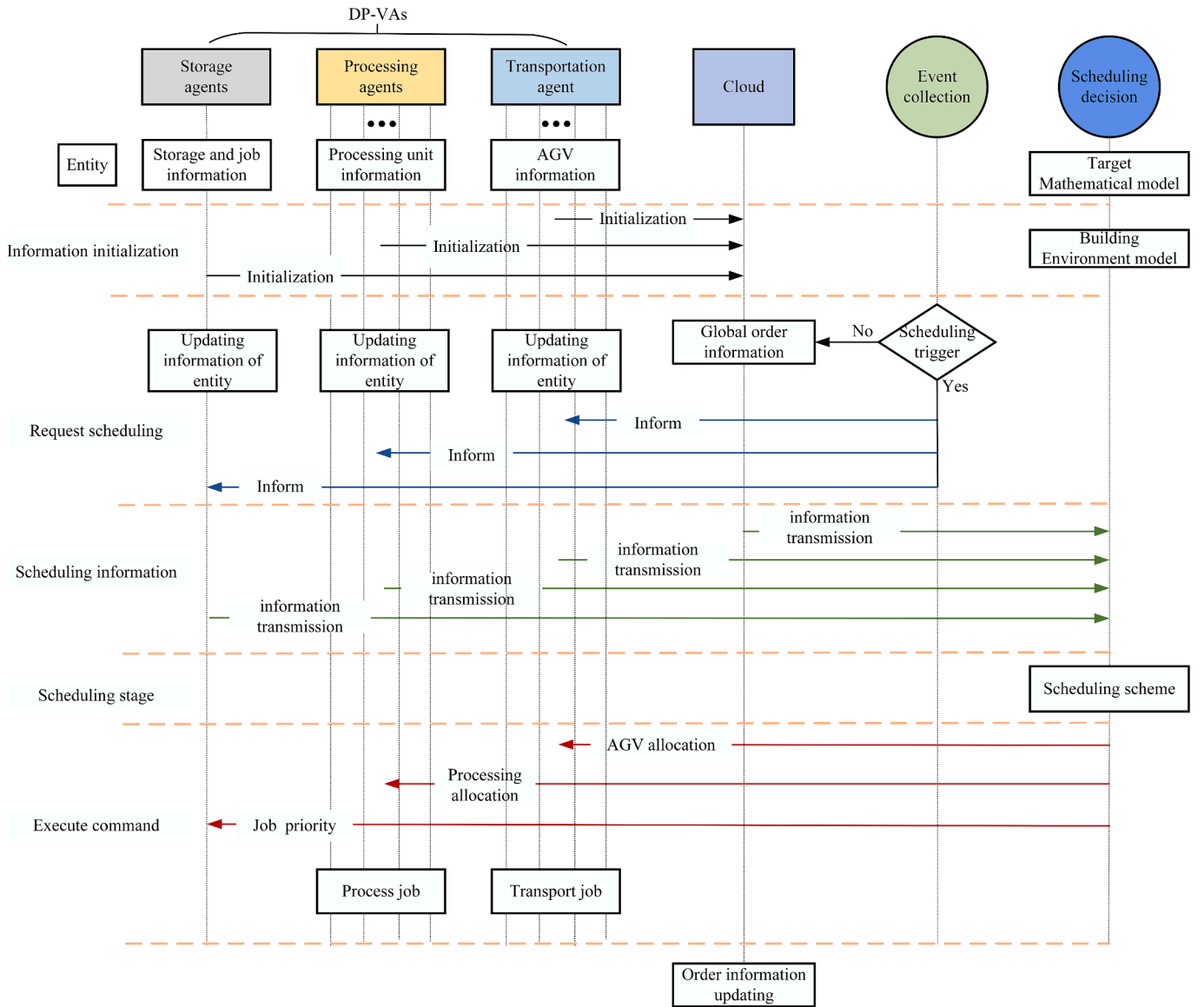


Fig. 7. The communication of mechanism.

5.3. Scheduling model

5.3.1. The scheduling framework

This paper adopts two-stage method as the scheduling framework. As shown in Fig. 8, the stage 1 is the training stage which is used to train the scheduling model in the AI layer. And the stage 2 is the texting stage which can text the performance of scheduling model. In the sage 1, training problems will be designed and imported to intelligent workshop as order. After an order arrived in the environment (workshop), the whole production will start processing. When the scheduling event occurs, the AI layer will collect the real-time data generated by intelligent workshop. These data will be calculated and then constructed into state space which is shown in Table 1. Neural network can choose suitable action according to state space. This paper adopts PPO algorithm to train the networks. There two networks in training stage which include actor network (AN) and critic network (CN). The AN is used to make an action in real-time according to environment. Then the CN will evaluate the action chosen by AN according to reward function which is obtained after the scheduling command is executed. It is noticed that action space consists of scheduling rules. Lastly, AN will adjust self-parameters according to the evaluation. When stage 1 is over, selecting the best AN

model as scheduling model. The AN model will be texted in stage 2. In stage 2, experimental problems will be generated randomly and sent to intelligent workshop.

5.3.2. The design of state space and action space

The above contents introduce environment model and production data which can be consisted of state space. In the Table 1, this paper introduces 26 production data. These data can represent the production status of intelligent workshop in real time. On the other words, when the environment changes, these data will also change accordingly. Therefore, this paper adopts the 26-production data as the state space.

This paper used improved genetic programming algorithm (IGP) to generate action space. The whole process can be seen in Fig. 9. This method is divided into three steps, including: (1) parameters initialization; (2) rule library initialization; (3) rule library evolution. In the first step, initialing the relevant parameters: population number, iteration number, maximum depth, crossover rate, mutation rate. And then, defining the function set (+, -, *, /, max, min) and terminal set, the terminal set consists of several common composite scheduling which is shown in Table 4. In the second step, using the mixing method (perfect method and growth method) to generate the initial population. Each

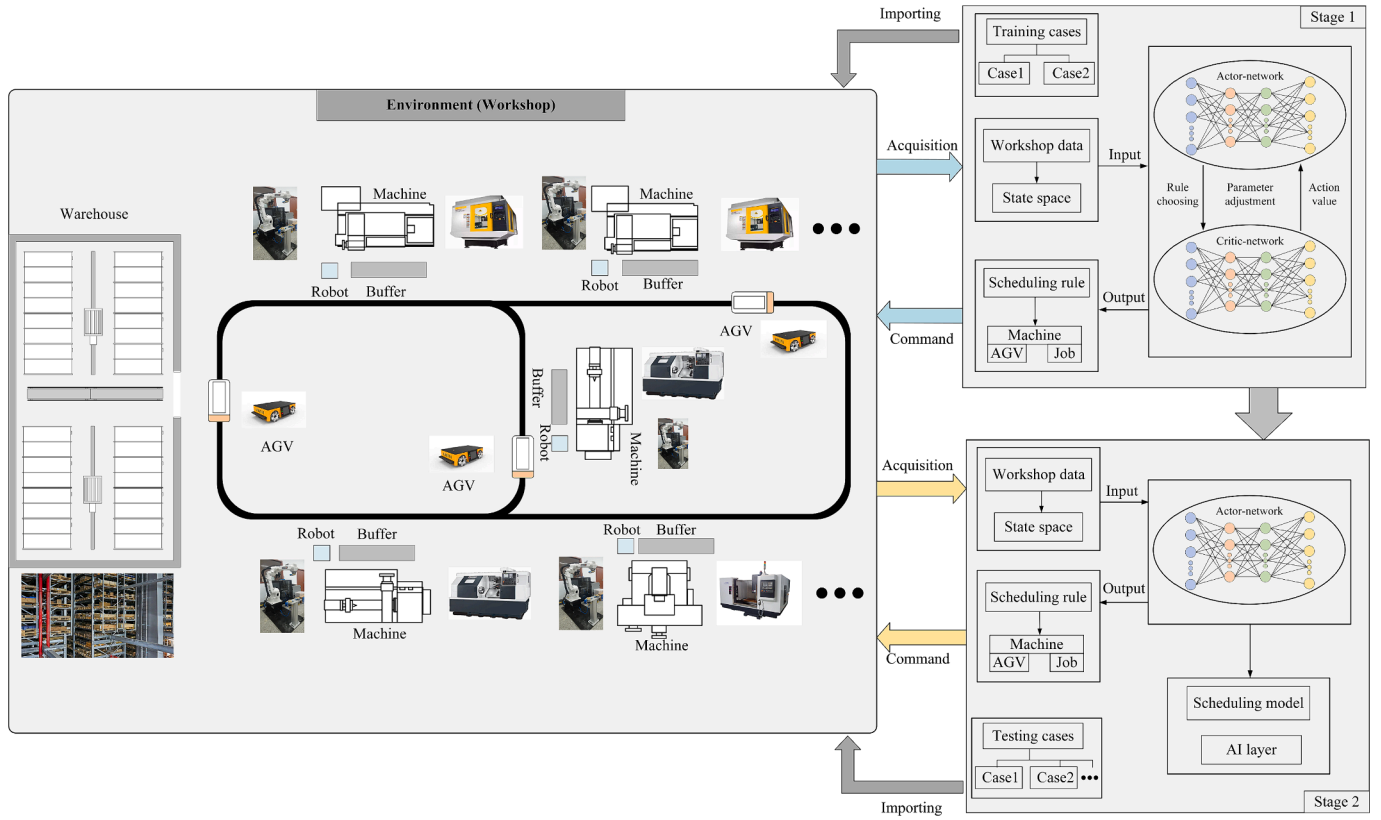


Fig. 8. Scheduling framework with two stages.

individual is a scheduling rule which represented by a binary try. Then, calculating each individual fitness value and selecting the best 12 individualities as the initial action space. Finishing the step 1 and 2, this method moves into step 3 and began iterative evolution. Firstly, the roulette method is used to choose the evolutionary population. This paper uses the improved crossover and mutation method to evaluate population. After finishing step 3, calculating the individual fitness value, and updating the action space. If the iteration is over, obtaining the best 12 scheduling rules as action space or repeating step 3.

The detail of improved crossover and improved mutation is described as follows. The traditional crossover method is to generate a random number within the population. Then cross the first half of the parent and the second half of the parent according to the generated random number. Based on the traditional method, this paper will calculate the fitness value of the progeny individual and then select the good individual as new individual through fitness value. And the improved crossover algorithms can be seen in algorithm 1. The traditional mutation method adopts single point variation. This paper uses the multipoint variation. Randomly selecting a certain range, and then mutation operation is carried out on all individuals with this range. The improved mutation algorithms can be seen in algorithm 2. And the benefits of improvement is described as follows: the common crossover algorithm generates a random number within the length range of the parent individuals. Then intercepting the first half of the father individual and the second half of the mother individual and cross the child individual based on the generated random number. The improvement of crossover is trying to cross genes within the individual length range of each parent, then calculating each fitness, and screen for the child individual with the highest fitness. This improvement can reduce the number of iterations and accelerate the convergence speed of fitness. The traditional mutation method adopts single point variation. This paper uses the multipoint variation and selective mutation probability which are described as follows: (1) Randomly selecting a certain range,

and then mutation operation is carried out on all individuals with this range. (2) By setting a small mutation rate and selectively mutating each gene on the child's individuality. In the other words, when the generated random number is less than the mutation rate, the gene mutates. When the traversal of gene positions is greater than half the length of the individuality, the mutation rate is set to twice the original. This improvement can effectively increase the frequency of individual updates, accelerate the rate of evolution and improve the quality of individuality.

Algorithm 1 Improved crossover algorithm

```

1: Function crossover (crossover rate, old individuals)
2: Calculate fitness of old individuals
3: Choosing best 12 individuals called excellent individuals (Ei)
4: X = number of old individuals, generate a parameter Cr randomly between [0,1]
5: For i = 1 to X do
6:   If Cr ≤ 0.9 then
7:     Execute crossover operation based on individuals and obtain current individuals
8:     Calculate fitness for each current individual (Ci)
9:     If Ci ≥ min (Ei) then
10:      new individual = current individual
11:     end if
12:   end if
13: next for
14: return new individuals
15: end function crossover

```

Algorithm 2: Improved mutation algorithm.

```

1: Function mutation (old individuals)
2: generate a mutation range M
3: define mutation rate = 0.35, X = population size
4: For i = 1 to X do
5:   mutation rate = 0.35, generate a number Mr randomly between [0,1]
6:   if i ≥ X/2 then

```

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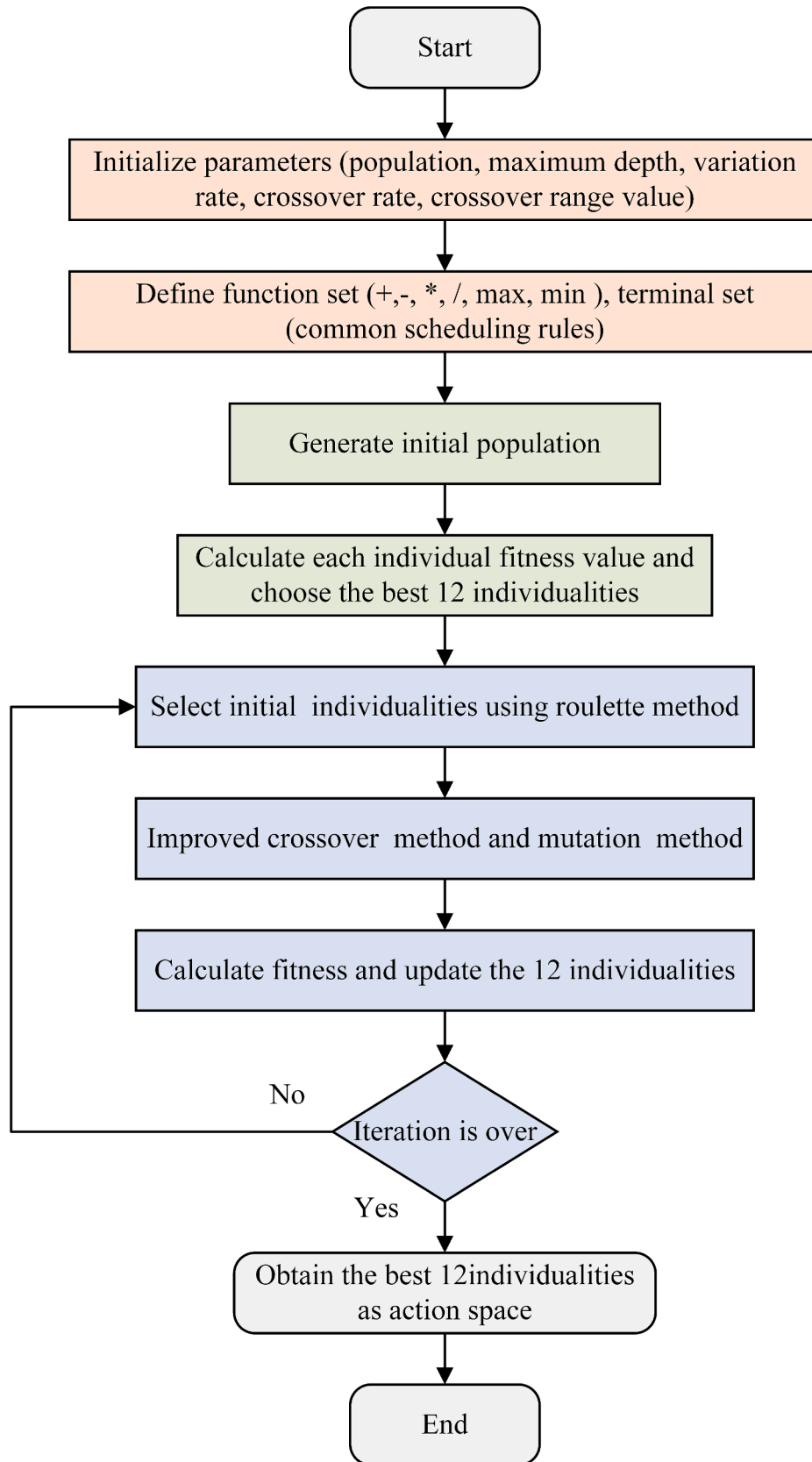


Fig. 9. The process of IGP to generate action space.

(continued)

```

7:      mutation rate = 0.7
8:      end if
6:      if Mr ≤ mutation rate then
5:      Execute mutation operations on all individuals within the range M
6:      obtain new individuals and update population
7:      end if
8:      next for
9:      return new individuals
10:     end function mutation

```

5.3.3. Training algorithms based on PPO

Reward function can guide the AI layer to make reasonable scheduling decisions and enable workshop scheduling to develop towards optimizing performance indicators. A reward function is designed according to scheduling goals. The reward function r consists of r_1 and r_2 . The r_1 is relevant to processing time and r_2 is relevant to energy consumption. And α and β represent weight ratios. The r which represents step reward is calculated by Eq. (13):

$$r = \alpha \cdot r_1 + \beta \cdot r_2 \quad (13)$$

The first reward r_1 is calculated by the single step time between two adjacent scheduling event trigger points. Defining a scheduling event trigger points as T_i , the calculating of r_1 is shown in Eq. (14):

$$r_1 = T_i - T_{i+1} \quad (14)$$

The second reward r_2 is relevant to energy consumption. From section 3, The energy consumption concludes processing energy consumption ME_p and transporting energy consumption TE_p . This paper calculates the energy consumption during a single time step. The r_2 can be calculated by Eq. (15):

$$r_2 = -(ME_p + TE_p) \quad (15)$$

In this way, we can calculate the step reward of production. Then, accumulating each step reward to obtain the total reward value. The reward value is related to the two scheduling goals.

Based on this, the training algorithm can be seen in algorithm 3. In line 1–4, the initialization condition and parameters are defined. Line 5–12 represent that production is executed in environment and relevant data is collected. In line 13–18, firstly, calculating advantages A_t which can not only compute the loss of critic network, but can be used in strategy gradient objective function. Then, the actor network parameter is updated by loss function $L_t(\theta)$. And $L_t(\theta)$ consists of three parts (strategy gradient objective function, value function objective function, entropy of action). After finishing an updating of network, the memory should be cleared. Lastly, the best actor network will be saved as the last scheduling for AI layer after the training is finished.

Table 4

The basic elements of the terminal set.

Type	Rule	Contents
MR1	MSPT	Choosing machine which has the shortest processing time
MR2	MNLR	Choosing machine which has the minimum load rate
MR3	MNEC	Choosing machine which has the minimal energy consumption.
MR4	MMEC	Choosing machine which has the machine with maximal energy consumption
MR5	MNBT	Choosing machine which has the minimal energy consumption
MR6	MLPT	Choosing machine which has the with the longest processing time
AR1	ASPT	Choosing AGV which has t shortest arriving time
JR1	JLRT	Choosing job with longest remaining time is selected first.
JR2	JSRT	Choosing job with shortest remaining time is selected first.
JR3	JLPT	Choosing operation of job with longest processing time
JR4	JSPT	Choosing operation of job with shortest processing time
JR5	JNEC	Choosing operation of job with minimal energy consumption
JR6	JMEC	Choosing operation of job with maximal energy consumption

Algorithm 3 Training algorithm based on PPO

```

1: Initialize parameters related to PPO
2: Initialize memory buffer D with batch size N
3: Define two neural networks Actor and Critic
4: Initialize total training episodes X
5: for episode = 1 to X do
6:   Reset environment of intelligent workshop
7:   for event point SP = 1 to M do
8:     Observe the state of the environment  $s_t$ 
9:     Use actor network to obtain the adoption probability of each action, and the appropriate rule is selected according to the policy distribution  $a_t$ 
10:    Executing the action and observe the new state  $s_{t+1}$ , obtain a reward  $r_t$ 
11:    One step scheduling experience ( $s_t, r_t, a_t, s_{t+1}$ ) is stored in memory D
12:  next for
13:  if the memory capacity is greater than the specified capacity then
14:    for Specified number of times do
15:      Randomly selecting a batch of historical decision-making experience from the memory
16:      Compute advantage estimates  $A_t$  based on the current value function  $V_t$  from critic network
17:      Calculate strategy gradient objective function  $L_t^{clp}(\theta)$ , value function objective function  $L_t^{VF}(\theta)$ , entropy of action  $S_\theta(s_t)$ .
18:      Using loss function  $\max(L_t(\theta) = -L_t^{clp}(\theta) + \alpha \cdot L_t^{VF}(\theta) - b \cdot S_\theta(s_t))$  to update the Actor and Critic network
19:    next for
20:    Clearing the memory
21:  end if
22: next for

```

6. Prototype system and comparison experiments

In this section, Case validation prototype experiment platform is constructed first to verify the feasibility of the proposed method. Then, the training and testing process will be introduced. To show the superior performance, the proposed method will be compared with other scheduling algorithms (common scheduling rules, GP scheduling rules, Q-learning). Lastly, the proposed method is trained and tested.

6.1. The environmental platform

Referring to the actual production workshop, this paper designs a case validation prototype experiment platform to verify the feasibility based on the proposed manufacturing system. The design of experimental platform can be seen in Fig. 10. As shown in Fig. 10, there are 9 machines and 4 AGVs in the platform. Machines is divided three types and can be used in three processing stages to process different operations. Each machine is equipped with a buffer zone and a robot ram. Robot arm, buffer, IPC and machine consist of a group of processing unit. And we use the local Ethernet network as the IoT network for the experimental platform to realizing the interconnection of various units. Automated Storage and Retrieval System (AS/RS) is not only the starting point of job processing, but also the processing ending point. We adopt UHF RFID tag to realize the transmission and tracking of workpiece information. And RFID tag has a 64 bits user area which is readable and writable to store the job information.

The proposed scheduling method has training stage and testing stage. In order to finish the training stage, this paper designs relevant instances. The detailed of these instances is introduced as follows: (1) Machine numbers: 9; (2) AGV number: 4; (3) job number and operation for each job: Unif [20,80], Unif [3,9]; (4) Processing time of each operation: Unif [15,35]; (5) The machine cutting power, machine idle power, AGV transporting power; Unif [4,12], Unif[{1,8 + 0.1*i, i = 1, ..., 19}], Unif [3.5,3.9]. The symbol Unif [a,b] represents uniform distribution from a to b; (6) Dynamic events: job insertion and machine fault. These can be seen in Table 5. Moreover, training stage is completed through simulation in simulation system.

6.2. The training and testing for IGP-PPO

Training stage is completed through simulation in simulation

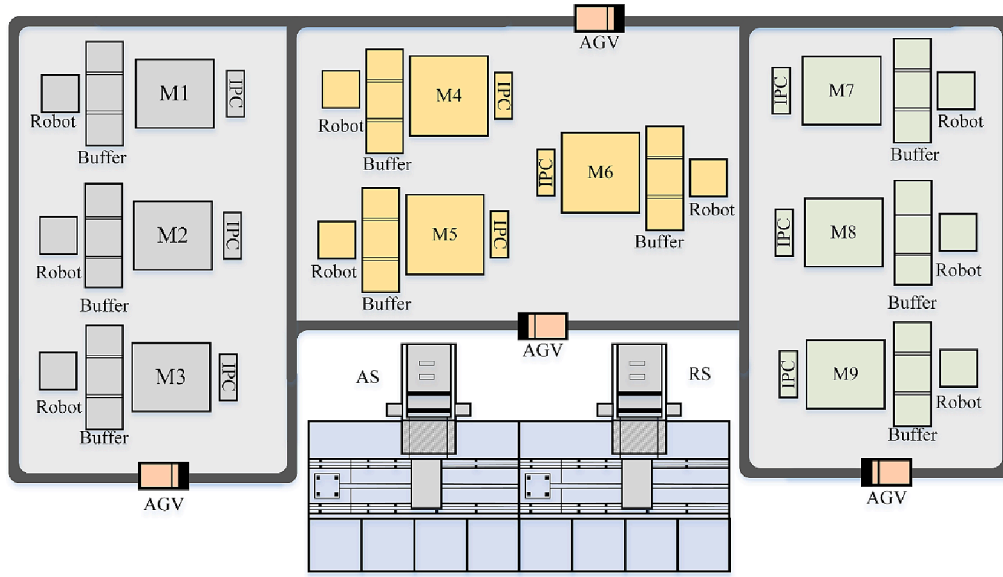


Fig. 10. The prototype experiment platform.

Table 5
The relevant parameters of instances.

Data	Values
Machine number	9
AGV number	4
Job and operation number	Unif [20,80], Unif [3, 9]
Processing time of each operation	Unif [15,35]
Machine cutting and idle power	Unif[4,12], Unif[{1.8 + 0.1*i, i = 1, ..., 19}]
AGV transporting power	Unif [3.5,3.9]
Dynamic events	Job insertion and machine fault

environment, which is configured as windows 10, 64-bit operating system, 16 G RAM, Intel i9 3.50 G Hz CPU. First, the IGP method is used to build action space which contain 12 high quality scheduling rules. This paper gives the parameters of IGP. The population number is set to 50, maximum depth is set to 3, crossover rate is 0.95, mutation rate is 0.05, mutation range is [10,25] and number of iterations is 80.

After obtaining action space, problems are imported into the environment to start training. The relevant parameters are given as follows: (1) Actor neural networks and Critic neural networks: fully connected mode; (2) Actor network structure: 26–128–128–32–32–8;(3) Critic network structure: 26–128–128–32–32–1;(4) Maximum number of epi-

sodes: 10000; (5);0.99; (6): Learning rate: 0.00065; (7) Memory bank capacity: 4096; (10) Batch memory size: 512; (11) $\lambda, \epsilon, c_1, c_2$: 0.98, 0.2, 1.0, 0.01. In the training process, each time the neural network model is updated, the corresponding parameters will be saved. The training result can be seen in Fig. 11.

Fig. 11(a) shows the relationship between the number of iterations and objective values. And Fig. 11(b) represents the relationship between the number of iterations and the mean value of the objective for each ten iterations. From Fig. 11, the curve of objective value decreases with the increase of iteration numbers. It means that AI layer can realize the learning to select suitable scheduling rules under various production conditions. Moreover, after the 9300 iterations, the training curve gradually converges without showing an obvious downward trend. Therefore, all the neural network models after the 9300th iteration are saved for comparison and selection. And the neural network which own the best scheduling performance will be chosen and put into testing stage. The selection and comparison process are shown in Fig. 11.

In the selection and comparison experiments, each neural network model is tested on 20 instances and the corresponding objective value is obtained. First, a is defined as the average of the objective value corresponding to all neural network models under an instance, and b is the objective value obtained by a neural network model under this instance.

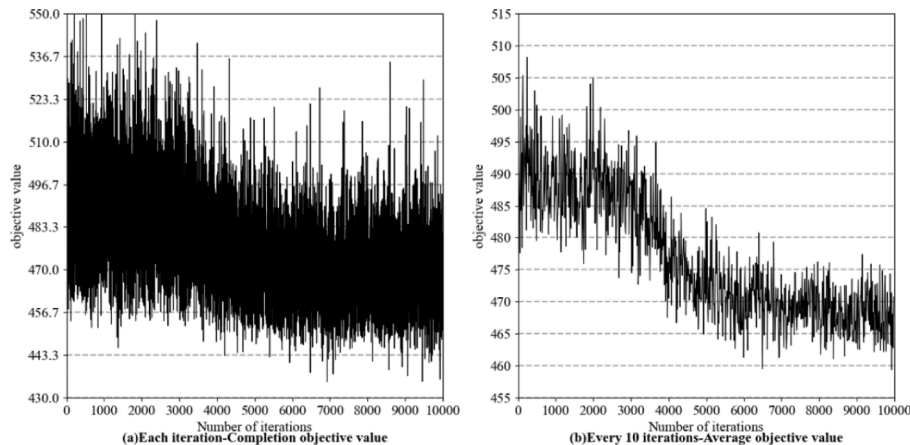


Fig. 11. The training process curve of IGP-PPO.

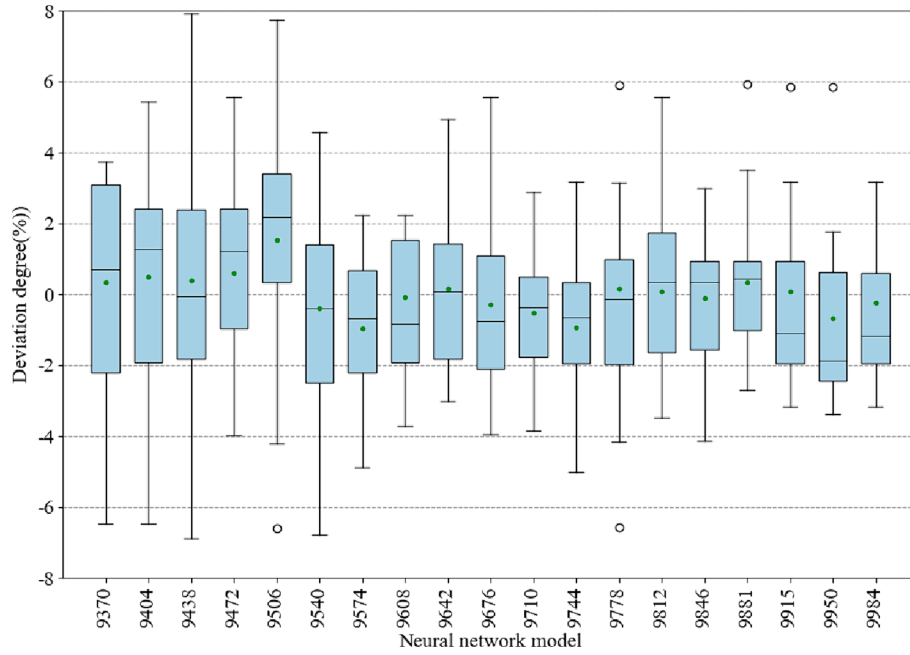


Fig. 12. The selection and comparison of neural network model.

Then, the deviation is equal to $(b-a)/a$. It can be seen that the smaller the deviation degree is, the better the performance of the neural network is. Fig. 12 represents a box diagram of the deviation degree of all saved neural network models for 20 validation instances. In the rectangles of each box diagram, the horizontal lines represent the median of each of the 20 degrees of deviation, and the solid points represent the corresponding mean. It is obvious that the neural network model saved in the 9574th iteration shows a smaller tendency in the deviation degree.

6.3. Comparative experiment

To prove the superiority of the proposed method, the comparative experiment is designed. In this experiment, the IGP-PPO is compared with three scheduling methods which include twelve composite scheduling rules, 8 well-known scheduling rules and GP rules. This paper designs 3

comparative experimental data sets and each sets have 20 problem instances. And parameter setting can be seen in Table 5. For scheduling performance comparison, superiority is defined as $(x_1 - x_2)/x_1$ here.

6.3.1. Comparison with composite scheduling rules

At first, the proposed method is compared with composite scheduling rules. These rules consist of three single rules from Table 3. These rules are given below: (1) MR3 + AR1 + JR2; (2) MR2 + AR1 + JR2; (3) MR1 + AR1 + JR1; (4) MR6 + AR1 + JR1; (5) MR5 + AR1 + JR1; (6) MR1 + AR1 + JR3; (7) MR5 + AR1 + JR3; (8) MR3 + AR1 + JR4; (9) MR4 + AR1 + JR4; (10) MR2 + AR1 + JR4; (11) MR5 + AR1 + JR5; (12) MR4 + AR1 + JR6. Then, comparative experimental data set 1 will be run under proposed method and using 12 composite scheduling rules. The results will be presented in the form of a box diagram which is shown in Fig. 13.

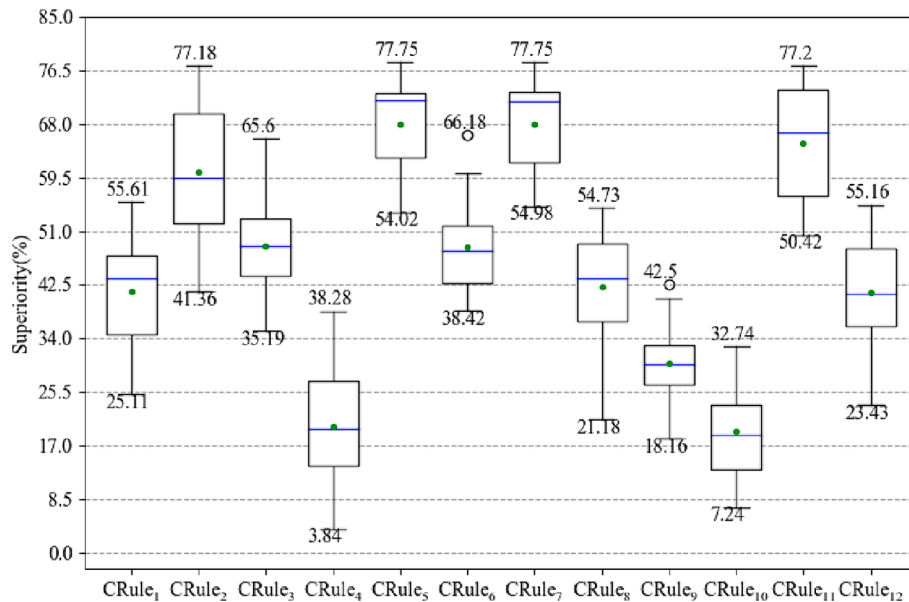


Fig. 13. The comparison with 12 composite scheduling rules.

Table 6

The basic elements of 8 widely used scheduling rules.

Rule	Contents
MSPT	Choosing machine which has the shortest processing time
MSFT	Choosing machine which has the shortest processing time in all buffers
MNEC	Choosing machine which has the minimal energy consumption.
MNBT	Choosing machine which has the minimal energy consumption
ASPT	Choosing AGV which has t shortest arriving time
JNER	Choosing the job with the minimal completion rate
JXCR	Choosing the job with the maximal completion rate
JLJT	Choosing the job with the longest total processing time
JSJT	Choosing the job with the shortest total processing time
JNOJ	Choosing operations with the smallest time
JMOJ	Choosing operations with the largest time
JLRT	Choosing job with longest remaining time is selected first.
JLPT	Choosing operation of job with longest processing time

For each box plot, the middle horizontal line and solid point are the median and average of the superiorities in 20 instances. And the upper and lower numbers represent the maximum and minimum values. From Fig. 13, it can be seen that the minimum value of each box plot is greater than 0, which means that the IGP-PPO achieves a smaller objective value in all instances. At the same time, the maximum value of some box diagrams can reach about 77 %, which means that the performance of IGP-PPO is far superior to the composite scheduling rules in some instances. In addition, the average values of 20 superior degrees corresponding to each box pattern diagram are 41.35 %, 60.34 %, 48.56 %, 20.03 %, 67.93 %, 48.42 %, 68.04 %, 42.18 %, 30.13 %, 19.2 %, 65.01 %, 41.24 %. Therefore, the IGP-PPO can obtain a smaller objective value and has better scheduling performance for FJSP-LTR compared with 12 composite scheduling rules.

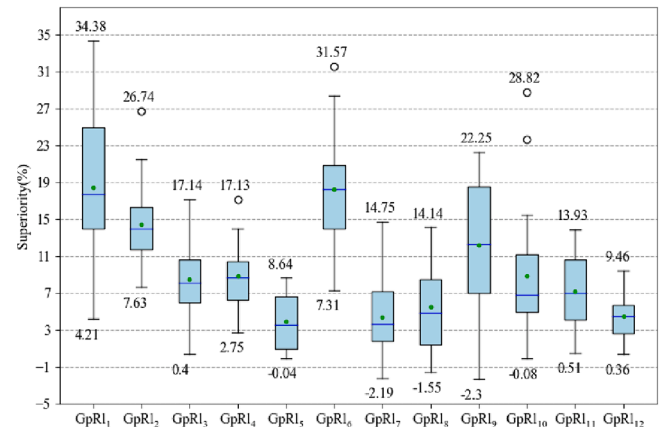
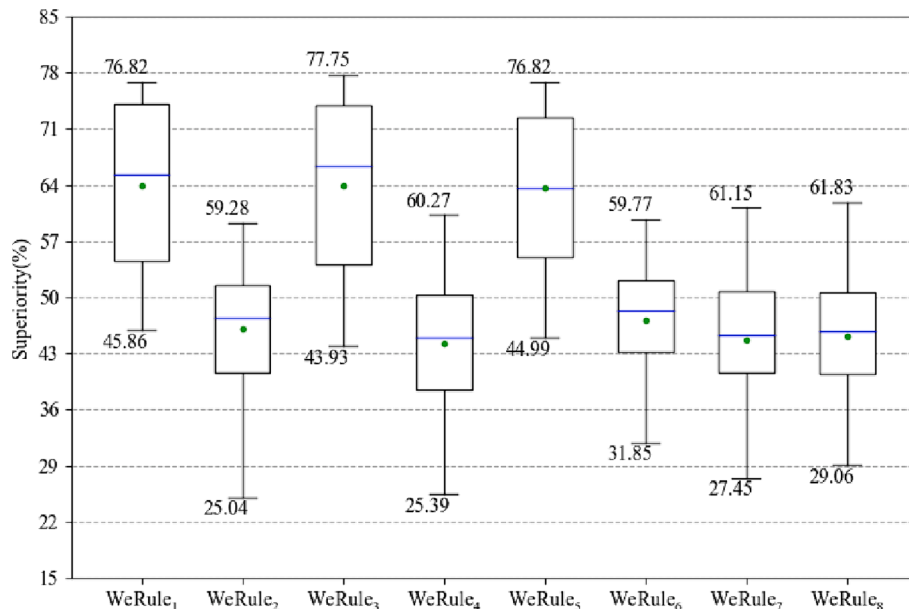
To further prove that the proposed method is better than a single scheduling rule, this paper selects 8 other common scheduling rules which is widely used in scheduling. And the element basic rules can be seen in Table 6. Based on this, the 8 widely used scheduling rules are given blow: (1) JNCR + ASPT + MSFT; (2) JXCR + ASPT + MSPT; (3) JLJT + ASPT + MNBT; (4) JSJT + ASPT + MNEC; (5) JNOJ + ASPT + MSFT; (6) JMOJ + ASPT + MSPT; (7) JLRT + ASPT + MNEC; (8) JLPT + A SPT + MNEC. The two methods will be tested independently under 20 instances, and results can be shown in Fig. 14. The average superiority values of IGP-PPO to the 8 scheduling rules are in 25 %-77 % respectively. Therefore, IGP-PPO can obtain lower objectives in all

instances, which can prove that the superiority of scheduling performance of proposed method in various scheduling problems compared 8 other scheduling rules.

6.3.2. Comparison with GP and RL

GP scheduling method is classic heuristic scheduling rules method. This paper firstly adopts IGP method to obtain action space. Therefore, the comparison between GP scheduling method and proposed method can directly improve the scheduling performance of IGP-PPO. This paper compares 12 GP scheduling rules which is generated by GP methods with the proposed method. The 20 instances are run under the two methods to obtain objective value. Then, calculating the superiority according to their objective value. The result can be seen in Fig. 15. The superior proportion of IGP-PPO to the 12 GP scheduling rules are in -2.3 %-35 % respectively. Although the value of superiority appears to be negative on some problems, the value of superiority is greater than 0 on most problems. Moreover, the average and median values for each box chart are greater than 0. The IGP-PPO method has better performance than GP rule method when dealing with different when solving scheduling problems of different scales.

Q-learning is a model-free, value-based RL algorithm designed to enable agents to make optimal decisions in uncertain environments. The

**Fig. 15.** The comparison with GP rules.**Fig. 14.** The comparison with 8 widely used scheduling rules.

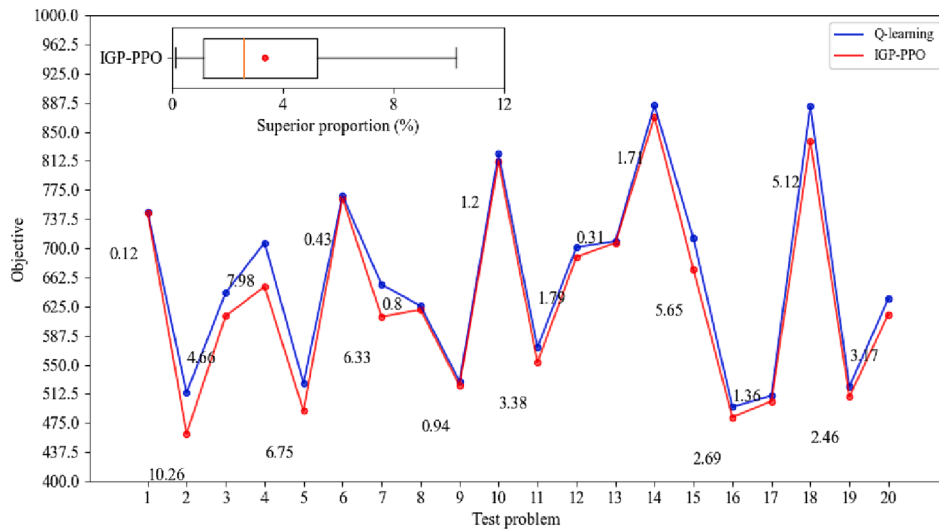


Fig. 16. The comparison with Q-learning.

core idea involves learning a value function (Q-function) for state-action pairs to guide the decision-making process of the agent. For the comparison experiment, we also use Q-learning method to solve scheduling problems. Firstly, designing x discrete action state for Q-learning. Then a Q table ($x \times y$) is built to store the Q values. Lastly, in order to obtain the suitable Q table, adopting Q-learning algorithm to train and verify. This paper text the performance of IGP-PPO and Q-learning by solving 20 instances in this comparative experiment. The broken line graph in the middle of Fig. 16 represents the target values obtained using two methods for 20 problems. And the box graph in the upper left corner of Fig. 16 represents the superiority of PPO over Q learning. The superior proportion are in 0%–12%. It can be shown that IGP-PPO method has a lower target value compared to Q-learning. When solving different scales of this scheduling problems, the proposed method has better scheduling performance and and robustness. Therefore, DRL has a better advantage than RL when solving scheduling problems in dynamic and complex environments.

7. Conclusion and future work

This paper presents a multi-agent manufacturing system based on IoT in intelligent workshop and adopts a dynamic scheduling mechanism based on DRL farmwork to solve scheduling problem of intelligent workshop. First, the architecture of multi-agent manufacturing is designed in detail. And taking FJSP as the specific production scenario, construct a mathematical model to describe the entire production process. Secondly, a DB-VPA which has information layer, software layer and physical layer is designed to realize information perception, communication interaction, device interconnection, and scheduling decision. The software layer contains AI layer, communication layer, and conversion layer. Physical layer includes equipment layer and job layer. Third, this paper proposes a dynamic scheduling mechanism which include environment building, communication mechanism and scheduling model for intelligent workshop. Specifically, this paper designs 26 production data to form state space and adopts IGP method to build action space. Based on this, scheduling model is trained through PPO. To verify the feasibility and superiority, a prototype experiment platform is designed. The training and texting experiment indicate that the proposed method can effectively addressing dynamic scheduling problems in intelligent workshop. Through the comparison experiments show that the proposed method has better scheduling performance.

Future work will focus on four aspects. (1) Using advanced data processing methods (data mining, data cleaning, data fusion, etc.) to

obtain more effective production data and explore the connection between different data. (2) Designing more reasonable action space and reward functions to improve the convergence, robustness and scheduling superiority of the algorithm. (3) We will further explore the workshop scheduling problem under dynamic problems. For example, dividing dynamic events into different levels of interference and designing different strategies to respond to corresponding interference events. (4) We will study the multi-agent reinforcement learning (MARL) method (MAPPO, MADQN, MADDPG) and adopts this method to solve complex and dynamic scheduling problems.

CRedit authorship contribution statement

Wenbin Gu: Writing – original draft, Supervision, Methodology, Funding acquisition, Conceptualization. **Siqi Liu:** Writing – original draft, Visualization, Validation, Software, Methodology. **Zhenyang Guo:** Writing – review & editing, Visualization, Validation, Formal analysis, Conceptualization. **Minghai Yuan:** Writing – review & editing, Funding acquisition, Formal analysis. **Fengque Pei:** Writing – review & editing, Software.

Data availability

Data will be made available on request.

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